QWeatherRiskF — Issue 3

Weather Feature Engineering & Anomaly Generation

Validation report based on crop_long_v1.csv. Weather measurements are deduplicated to district-year records; district-specific normals are calculated independently within each leave-one-year-out (LOYO) fold; anomalies and provisional stress indicators are generated using those fold-specific normals.

Dataset and output summary

Measure	Result
Source crop-long rows	1,430
Unique district-year observations	65 (13 districts × 5 years)
Weather features	78 (13 indicators × 6 seasonal windows)
Missing values in unique district-year weather data	169
District weather normal rows	5,070
Weather anomaly rows	25,350 (5 folds)
Weather stress-feature rows	325 (5 folds)

Fold-safe methodology

For each held-out validation year, the script calculates district/window/indicator means and sample standard deviations using only the other four years. That fold-specific normal is applied to all observations in the fold, including training and held-out rows. The held-out year's weather is excluded from its fold's normal.

Absolute anomaly = actual − training mean.
Percentage anomaly = (actual − training mean) / training mean × 100, when the mean is non-zero.
Standardised anomaly (z-score) = absolute anomaly / training sample standard deviation, when the standard deviation is positive.

Independent validation recomputes each fold normal directly from the raw observation table after excluding the held-out year, then compares the mean, sample standard deviation, and valid training count with all generated normal rows. This verifies the numerical normal values, in addition to checking the recorded training-year exclusions.

Validation checks — 11/11 passed

Validation check	Result	Detail
Crop-long source key uniqueness	**PASS**	0 duplicate source keys
Weather values consistent across crop rows	**PASS**	0 district-year/weather-column combinations differ
One weather record per district-year	**PASS**	65 unique district-year rows
No held-out year enters its fold's training normals	**PASS**	Verified training_years for every held-out-year fold
Independent recomputation of fold normals	**PASS**	Means, sample standard deviations and valid training counts independently verified for all normal rows
Expected number of weather normal records	**PASS**	5,070 rows; expected 5,070
Expected number of weather anomaly records	**PASS**	25,350 rows; expected 25,350
Expected number of weather stress records	**PASS**	325 rows; expected 325
Missing weather values remain missing	**PASS**	845 missing fold-observation values; expected 845
Weather feature count is 78	**PASS**	78 features; expected 78
Fold-specific normals remain consistent	**PASS**	Each district-feature group uses one normal throughout each fold

The checks verify source-key and weather consistency, output dimensions, missing-value preservation, fold exclusion, independently recomputed normals, and within-fold normal consistency. Results are recorded in validation_tables/validation_checks.csv.

Missingness finding

The source contains 169 missing weather values across the unique district-year observations. All occur in **rabi_prev** for 2015 (13 indicators × 13 districts). Missing observations are retained; no zero-filling, interpolation, or imputation is performed. Across five folds, these produce 845 missing fold-observation values in the anomaly output.

Provisional stress screening rules

Feature	Rule	Interpretation
Rainfall deficit	rain_sum percentage anomaly ≤ –20%	Provisional screening heuristic
Rainfall excess	rain_sum percentage anomaly ≥ +20%	Provisional screening heuristic
Extreme rainfall	max_5day_rain z-score ≥ +2	Statistical proxy, not an official extreme-rain definition
Dry-spell stress	longest_dry z-score ≥ +1.5	Higher-than-normal longest dry spell
Heat-stress screening	tmax_mean or hot_days z-score ≥ +1.5	Provisional statistical proxy

These thresholds are initial statistical screening heuristics, not official meteorological definitions or crop-specific agronomic thresholds. They require review and calibration by the project/domain team for crop, growth stage, location, and source units. A missing anomaly or z-score produces a missing flag, not a false flag.

Limitations and handoff notes

1. Only five calendar years (2015–2019) are available. These are short-period baselines, not standard 30-year climatological normals.

2. The source CSV does not include an explicit weather-unit dictionary. Confirm authoritative source units and definitions before scientific interpretation or publishing unit labels.

3. LOYO folds use the other four years, including years later than the held-out year. Evaluate forward-chaining if the modelling question is strictly prospective forecasting.

4. The ne and kharif windows may overlap according to project dataset documentation. Do not aggregate them as independent periods without resolving the overlap.

5. Stress thresholds require domain review and calibration; they should not yet be treated as crop-specific decision rules.

Conclusion

The latest reported execution generated the expected output dimensions and passed all 11 scripted structural and data-quality checks, including independent recomputation of fold-specific normal means, sample standard deviations, and valid training counts. Outputs are available for team review, subject to confirmation of source weather units and domain review of provisional stress thresholds.